Asya Ünal

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Expected graduation: June 2027 · Available full time from Feb 2027, incl. Summer 2027 (13 to 17 weeks)

EDUCATION

Bilkent University, B.S. Computer Science — Ankara · *Sep 2023 – Jun 2027*
GPA 3.65/4.00 · High Honor every semester · Coursework: Data Structures, Algorithms, OS, Databases, Networks
City University of Hong Kong, Exchange Semester (Computer Vision, AI, Signals & Systems) — *Jan – May 2026*
Ankara University, Faculty of Medicine, completed three years before switching to CS — *Sep 2020 – Sep 2023*

EXPERIENCE

AlgoritiX (CodeAPeel) | *Programmer, part time, with Prof. Y. Oruç (UMD)* — Remote · *Aug 2026 – Present*

- Built a **FastAPI** AI assistant that runs model-generated assembly in a bounded sandbox inside an execute, observe, repair loop, so only programs that pass verification are proposed as editor changes
- Lead development of a React/TypeScript workspace for 4 instruction sets (MIPS, RISC-V, VESP, Ark-U) with execution, register and memory inspection and test-backed challenges; maintain the Java desktop tools

Mobile Action | *Software Engineering Intern* — Ankara · *Jun – Jul 2025*

- Proposed and built, unprompted, a Model Context Protocol server exposing **27** App Store analytics endpoints as LLM agent tools; demoed it to the engineering team
- Shipped SearchAds features end to end with Vue 3, Spring Boot and RabbitMQ

Eluvium | *AI Engineer Intern* — London · *May – Jun 2025*

- Re-architected a monolithic state machine into a modular **LangGraph** workflow in a multi-service production codebase
- Built LLM agent components on Gemini, including a **LangChain** classifier that routes manufacturing alerts to the right handler and reduced manual triage

Teknopar Industrial Automation | *Machine Learning Intern* — Ankara · *Jul – Aug 2024*

- **98%** accuracy on metal defect inspection with CNN and YOLO models (first-author, IEEE FiCloud 2025); DistilBERT on 3K+ error logs at **85%**

RESEARCH

KUIS AI Center, Koç Univ. | *Undergraduate Researcher, Prof. A. M. Tekalp* — Istanbul · *Sep 2026 – Present*

- Fine-tuned **Qwen2.5-VL-3B** with LoRA as a reference-free judge of diffusion super-resolution outputs; built resumable PyTorch train/eval pipelines with leakage-safe splits and hash-verified artifacts

Bilkent University, Dept. of EEE | *Undergraduate Researcher, Prof. S. S. Kozat* — Ankara · *Sep 2026 – Present*

- Agentic LLM orchestration for drone swarms, replanning under GPS or comms loss

PROJECTS

Miper [miper.ai] | *Founder and sole engineer* — *2025 – Present*

Architecting an **agentic AI platform** where specialist agents reason over a person's knowledge graph to plan, track and complete their work.

- Designed a **shared-memory multi-agent architecture**: journals, voice, photos, email and documents flow into one knowledge graph with vector search and community detection, and each agent acts on it through its own tools, model and proactive triggers
- Engineered a **hallucination-resistant ingestion layer**: a two-stage LLM cascade extracts commitments, deadlines and meetings, admits only facts anchored to exact source spans, and reconciles updates that supersede or contradict earlier ones
- Built the full stack solo: FastAPI/LangGraph backend, queue workers, Postgres/pgvector, React and React Native

Qardex [qardex.app] | *Founder* — *2026 – Present*

- Flutter/Firebase app turning notes into AI flashcards and real-time multiplayer study races

PUBLICATIONS

A. Ünal et al., AI-Driven Collaborative Assembly Line Inspection: System Integration and Deployment Challenges, *MobiWIS 2026*. arXiv:2609.33522

A. Ünal et al., Surface and Edge Defect Inspections on Metals Using Deep Learning and Optical Camera Data, *IEEE FiCloud 2025*.

F. Kaya, **A. Ünal** et al., Real-Time Anomaly Detection in Industry 4.0 Using Asset Administration Shell, *Springer LNCS 16066*, 2025.

HONORS & LEADERSHIP

- Ranked **2,670th of 2.4 million** (top 0.11%) in Turkey's national university exam; Bilkent Merit Scholarship, raised from 40% to 60%
- **ACM Bilkent** President 2024–25 (400+ members): grew event participation from **207 to 911** in a year; ran the ALPOC programming contest (154 teams, 26 universities); now Head of Audit

SKILLS

Languages: Python, C++, Java, TypeScript/JavaScript, C, Dart, SQL **Spoken:** Turkish, English (C2), German (B1)
Backend & Mobile: FastAPI, Spring Boot, PostgreSQL, RabbitMQ, Docker, Linux, Git, Flutter, React Native, React
ML & AI: PyTorch, Hugging Face, LoRA, VLMs, diffusion models, LLM agents (LangGraph, MCP), RAG, NLP